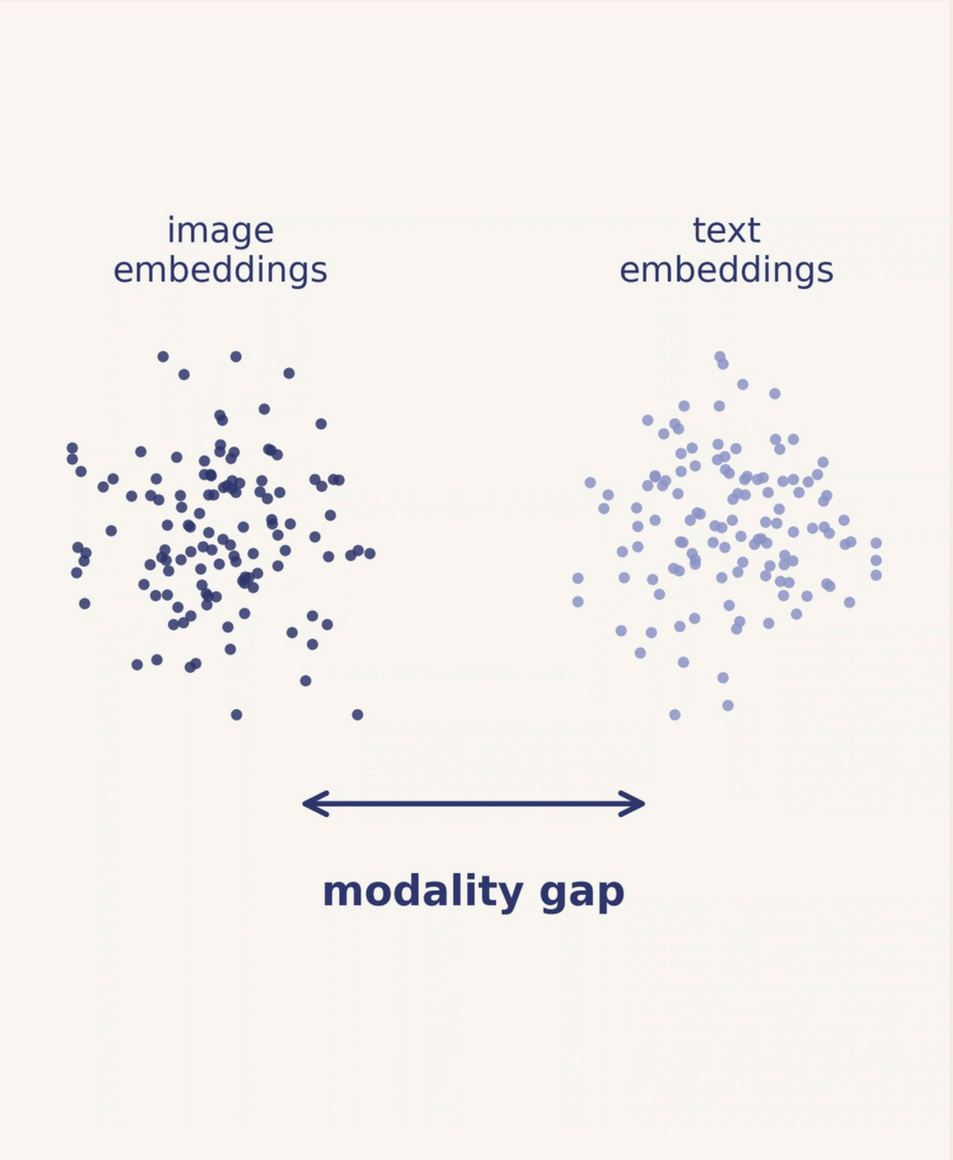
PROBLEM & MOTIVATION

Does closing the modality gap help hard-negative-trained CLIP?

CLIP embeds images and text in separate regions of one space, known as the modality gap. Hard-negative fine-tuning (NegCLIP) improves compositionality, but nobody has asked how the gap interacts with it. Recent work shows closing the gap makes CLIP more robust without hurting accuracy. We test whether that carries over to hard-negative-trained models.



RELEVANT WORK

Mind the Gap

Liang et al., NeurIPS 2022. Identifies a persistent separation between image and text embeddings in CLIP. The paper attributes the gap to a cone effect from model initialization and to the contrastive-learning objective, which preserves the separation during training. It also shows that changing the gap can affect downstream zero-shot performance and fairness.

Bug or Feature?

Chowers et al., 2026. Shows that, under certain conditions, the image-text separation can be represented by a global gap vector that is orthogonal to the embeddings. Their post-processing experiments move one modality toward the other and find that reducing the gap can improve robustness to embedding perturbations without changing clean accuracy.

Hard Positive Truth

Kamath et al., ECCV 2024. Shows that hard-negative fine-tuning can improve standard compositionality benchmarks while hurting performance on semantically related hard-positive captions. On their hard-positive evaluation, hard-negative-tuned CLIP can lose up to 38.7% performance. The work motivates checking robustness and compositionality beyond hard-negative benchmarks alone.

PLANNED APPROACH

01 Measure the gap

Compute the image-text mean offset for OpenCLIP and NegCLIP (inference only, baselines).

02 Alter the gap

Orthogonal shift of text embeddings along the gap vector, plus a mean/median shift by a set margin.

03 Add a control

OpenCLIP trained without hard negatives, to isolate the effect of hard-negative training.

04 Evaluate

Run every gap setting on CLIP_benchmark compositionality tasks and an image-captioning dataset.

EXPECTED DELIVERABLES

Mid-project presentation

- Modality gap measured for OpenCLIP and NegCLIP
- Orthogonal shift implemented, with first results on baseline OpenCLIP
- Evaluation pipeline running on a compositionality benchmark

Final presentation

- Results for NegCLIP vs. the OpenCLIP control across gap sizes and shift types
- A clear answer: does closing the gap help hard-negative-trained CLIP?
- Analysis of accuracy and robustness trade-offs

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